MAKING AN IMPACT WITH DATA:

UNCOVERING DEPAUL'S GLOBAL STUDENT PIPELINE

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Tools Used: Power BI, Excel, Python

Excelerate

Data-Driven Analysis of International Student Admissions at DePaul University

This project focuses on analyzing international student admission data from DePaul University to understand application trends, student demographics, and program preferences. Using data analysis and an interactive dashboard, the study provides insights to improve recruitment strategies and support data-driven decision-making.

Key Skills Required

Skill 01

1. Data Cleaning & Preparation
   - Handling missing values, duplicates
   - Managing inconsistent data (merged systems, duplicate columns)

👉 Example: Cleaning "University, University-2, University-3" columns

Skill 02

2. Data Analysis & EDA
   - Understanding trends and patterns
   - Country-wise, program-wise analysis
   - Identifying key insights

Skill 03

3. Technical Skills
   - Excel: Data cleaning, basic analysis
   - Power BI: Dashboard creation, DAX
   - Python (optional): Data manipulation (Pandas)

Skill 04

4. Data Visualization
   - Creating charts, KPI cards
   - Designing user-friendly dashboards
   - Making data easy to understand

Skill 05

5. Analytical Thinking
   - Converting raw data into insights
   - Finding problems (e.g., student drop-off points)
   - Suggesting solutions

Understanding Data

Structured Data

To make it structured, the data was cleaned and organized by removing duplicate fields, handling missing values, and standardizing formats. Relevant information was then arranged into clear columns like country, program, intake, and admission status, converting the dataset into a consistent tabular form suitable for analysis and dashboard creation.

Unstructured Data

The DePaul University dataset was initially in an unstructured and inconsistent form because it was merged from multiple systems. This resulted in duplicated columns (like University, University-2, University-3), mixed formats, and scattered information such as student profiles and interaction logs, making it difficult to analyze directly.

Data Cleaning and Preparation

Before vs After Summary		
What We Measured	**Before Cleaning**	**After Cleaning**
Total Columns	80	36
Total Rows	7,543	7,543
Duplicate Records	2,543	0
Unique Applicants	Unknown	5,000
Total Missing Values	371,657	Only 802 (college_degree & program)
Date Format	Integer numbers	Proper date-time
College Info	1 merged column	3 clean separate columns
'Unknown' placeholders	Scattered everywhere	Replaced with proper null

Tools Used in Week 2:

1) Python (Pandas) — for all cleaning steps, column operations, interapolation method , and text fixing.
2) Excel / CSV — for visual inspection before and after.
3) Google Sheets — for documenting the data cleaning log and comparison summary.

Exploratory Data Analysis (EDA)

The first step of this project was to understand the raw dataset before touching anything. We received a DePaul University admissions export containing 7,543 records and 80 columns covering student demographics, program preferences, application status, and communication history.

During EDA, we profiled the entire dataset to assess its structure and quality. We quickly discovered that the data was far from analysis-ready — over 30 columns were completely empty, the supposed unique identifier (Reference_ID) was heavily duplicated with one ID appearing up to 48 times, timestamps were stored as unreadable integers, and text fields like country names had inconsistent formatting throughout.

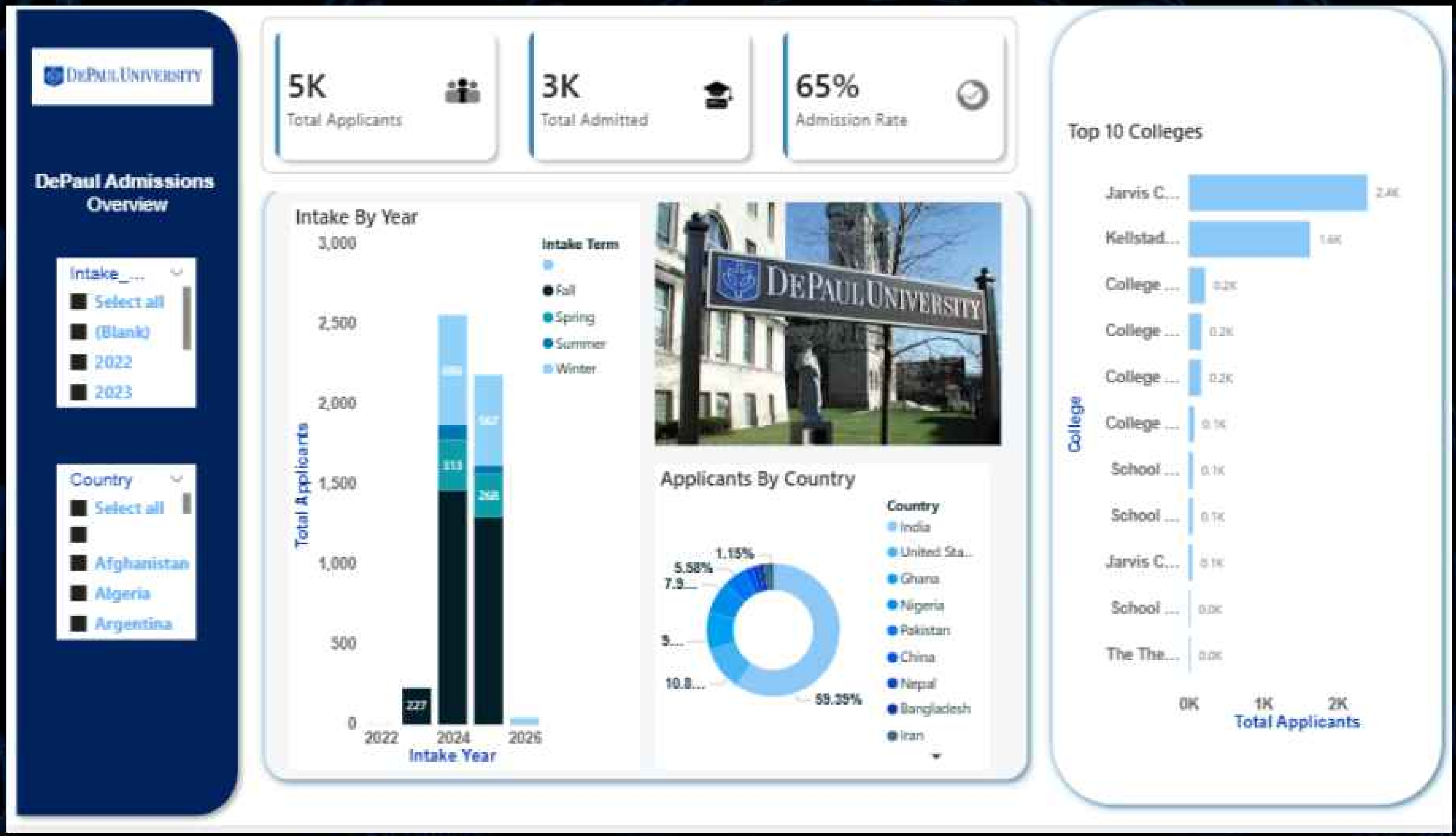
We also ran initial visualizations to understand patterns in the data. These revealed that India dominated the applicant pool at 59%, Computer Science was the most popular program, and student intake was heavily concentrated in the Fall period. A data dictionary was built alongside this exploration to document every variable and ensure the whole team understood the dataset the same way.

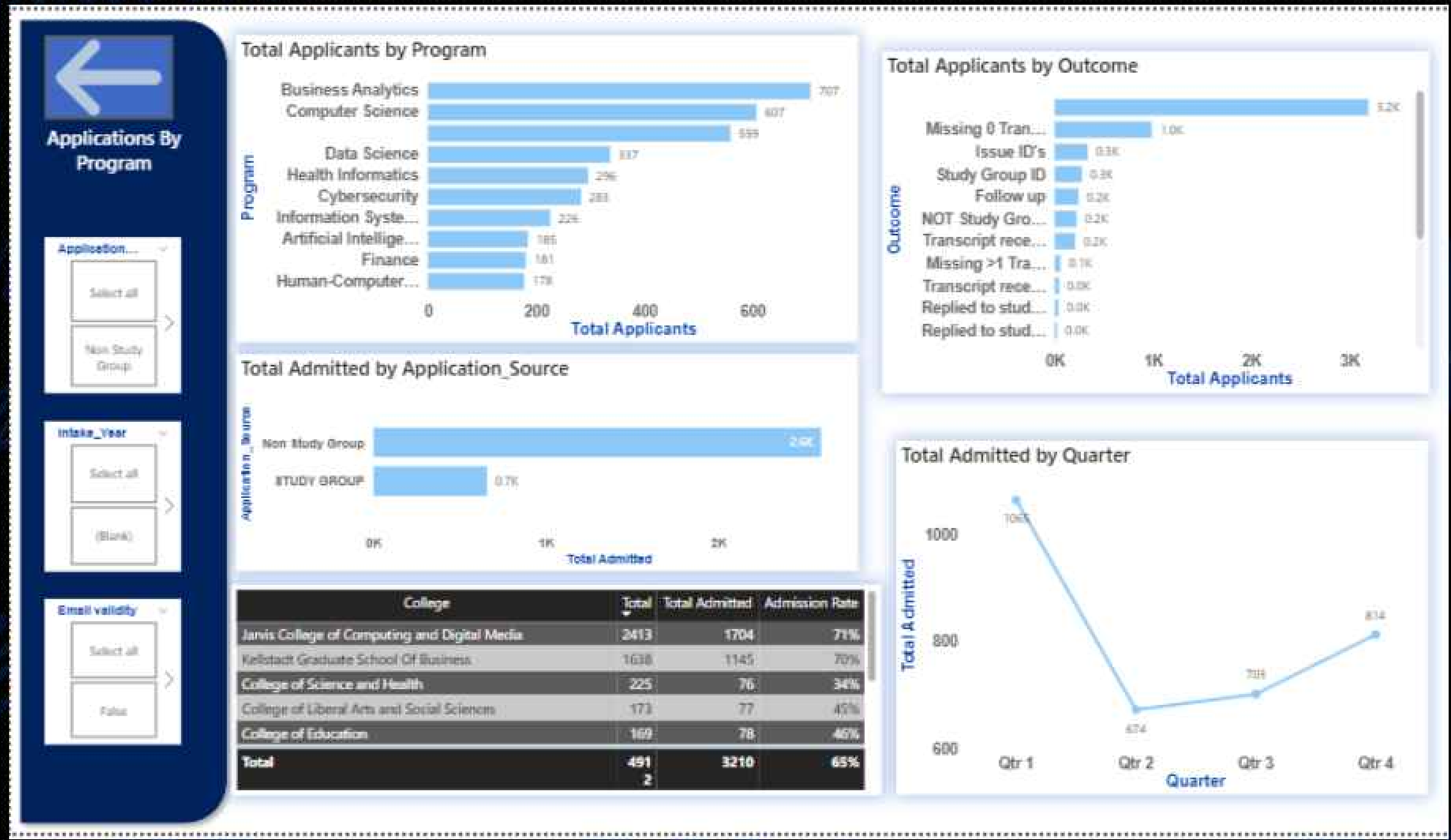
The overall conclusion of EDA was clear — the dataset had serious structural and quality issues that made it unreliable for any direct analysis. Every finding was documented and handed over to the cleaning phase in Week 2, where the real transformation began.

Statistical Analysis Techniques

S.No.	Technique	What We Did	What We Found	Measure Used
1	**Frequency Distribution**	Counted how many times each value appeared in columns like Country,	India appeared 4,488 times — the single largest group in the	**Count per category**
2	**Descriptive Statistics**	Calculated mean, median, minimum, and maximum values for all numerical columns	Reference_ID had a mean of ~473M — confirming it was just a system ID with no	**Mean · Median · Min · Max**
3	**Missing Value Analysis**	Calculated the number and percentage of empty or null entries across all 80 columns	Remark_2 was 99% empty · 30+ columns had zero data · 371,657 total missing values	**Null count · % missing**
4	**Duplicate Detection**	Counted how many times each Reference_ID appeared to find repeated student records	2,543 duplicate rows found · one ID appeared 48 times · only 5,000 truly	**Duplicate row count**
5	**Percentage Calculation**	Converted raw counts into percentages to make patterns easier to understand and present	India = 59% · Computer Science = most popular · Fall = highest intake period	**Proportion · Share %**
6	**Unique Value Count**	Checked how many distinct values existed in each column to understand data diversity	Country had 95 unique values · Program had 19 · helped spot	**Cardinality · Distinct count**
7	**Distribution Check**	Looked at how values were spread across categories using bar charts and frequency	Data was heavily skewed — a few countries and programs dominated	**Spread · Skewness**
8	**Data Profiling**	Examined each column's data type, sample values, and structure to spot formatting problems	Dates stored as integers · merged columns · "Unknown" used as filler instead	**Data type · Format check**

DePaul University Admission Analytics





Key Insights

Dataset Insights

Student Demographics & Geography:

- Most applicants originate from India, making it the dominant feeder country
- Growing contributions from emerging markets like Ghana and Nigeria
- Heavy reliance on a few key feeder institutions, indicating limited diversity in recruitment sources.

Program Preferences:

- Strong student preference for STEM programs, particularly:
- Computer Science
- Data Science
- Less popular programs receive significantly fewer applications.

Application & Admission Trends:

- Fall intake receives the highest volume of applications compared to other intakes
- Key funnel metrics tracked: Total Applicants, Total Admitted, and Admission Rate (%)
- The admission rate reflects the balance between institutional demand and selectivity

Recommendations from the Dashboard Report

Recommendations:

- Diversify recruitment geographically — expand outreach to new regions beyond the current dominant sources to broaden the student base
- Strengthen feeder institution relationships — reduce over-reliance on a few institutions by identifying and engaging new partner schools
- Prioritize Fall intake resources — since Fall dominates applications, concentrate recruitment efforts around that cycle
- Promote underrepresented programs — run targeted campaigns for less popular (non-STEM) programs to balance enrollment
- Leverage interaction insights — use dashboard engagement data to refine and personalize student engagement strategies

We didn't just analyze the data — we made it explorable. Click the link [Presentation Link](#) on this slide and you can filter by country, program, or intake period yourself and discover the story the data tells.

Conclusion

Through this project, we successfully transformed unstructured admission data into a structured format and analyzed it to uncover meaningful insights. We identified key trends in student demographics, program preferences, and application patterns, with a strong demand for Science, Technology, Engineering, and Mathematics programs and higher applications during the Fall intake.

The interactive dashboard helped us visualize these insights clearly and made data-driven decision-making easier. Overall, this project enhanced our skills in data cleaning, analysis, visualization, and teamwork, while also providing valuable recommendations to improve recruitment strategies.

THANK YOU!

WE APPRECIATE YOUR VALUABLE TIME.